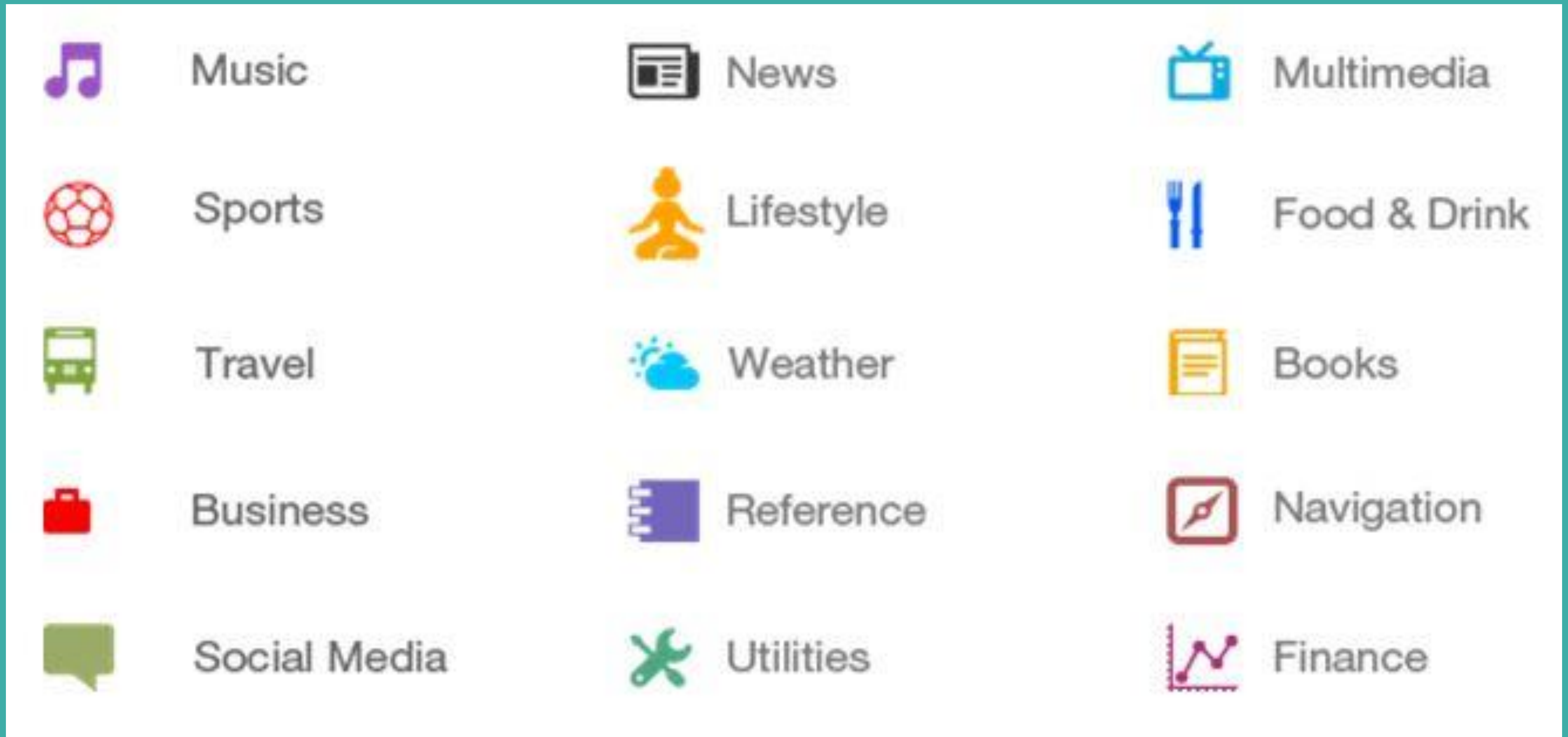
Android Programming

Why Mobile Programming

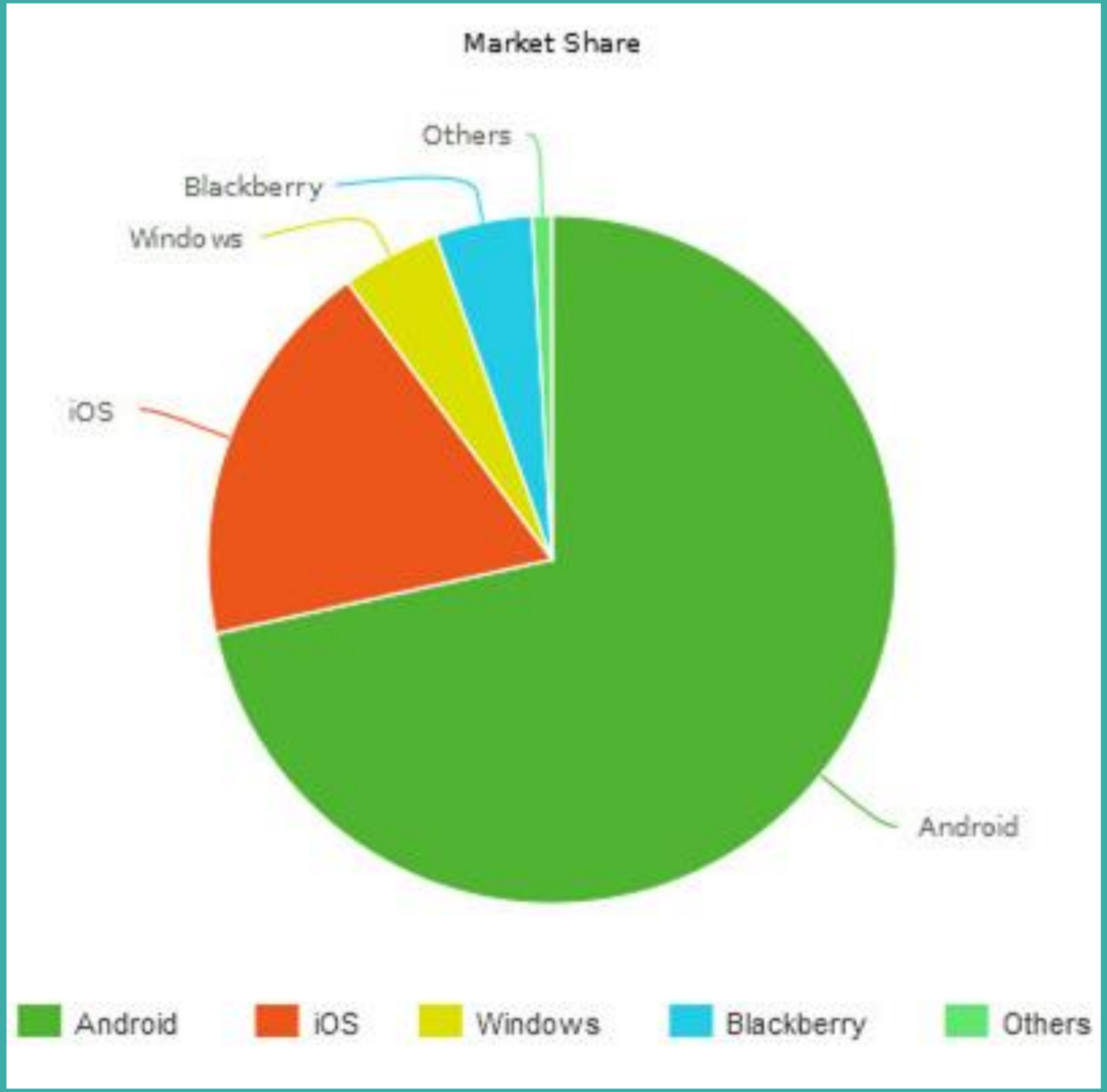
Just think what do you do most in a day

- Mobile apps are ruling the world
- Good hardware and software support
- Convenient than other means
- Gather real time data with built-in sensors
- Reaching out to all age of users

Mobile Applications for Everything

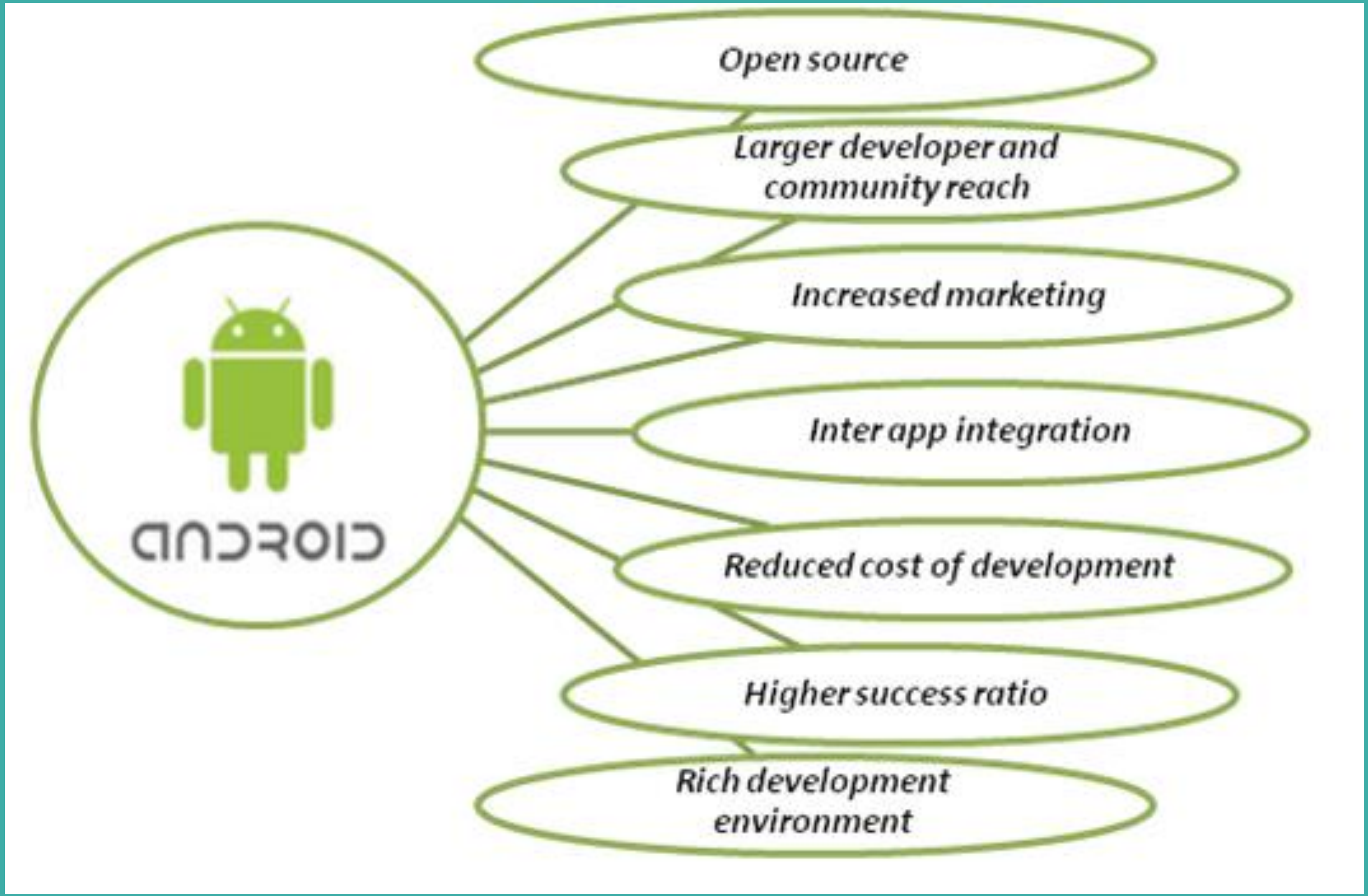


Mobile Applications Development



*https://www.androidauthority.com/develop-apps-for-android-rather-than-ios-607219/

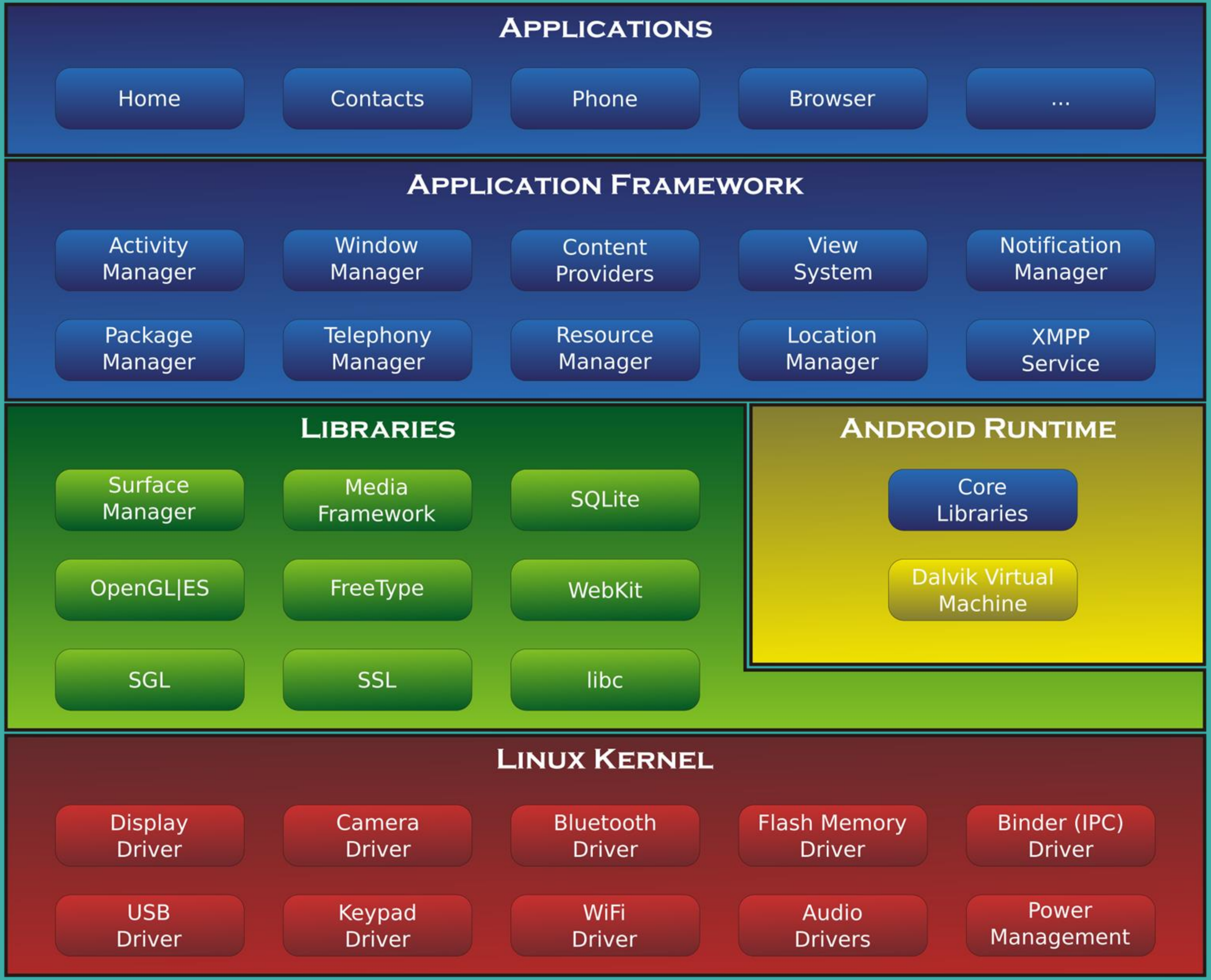
What is Android



- Linux based OS
- Open Source
- Can develop app for mobile devices

History of Android

System Architecture



Android Programming Platforms

- **MIT App Inventor (Blocks)**
- Android Studio
- Unity 3D
- B4A (Basic for Android)

MIT APP Inventor

MIT App Inventor

Laptop or Desktop

Gmail account

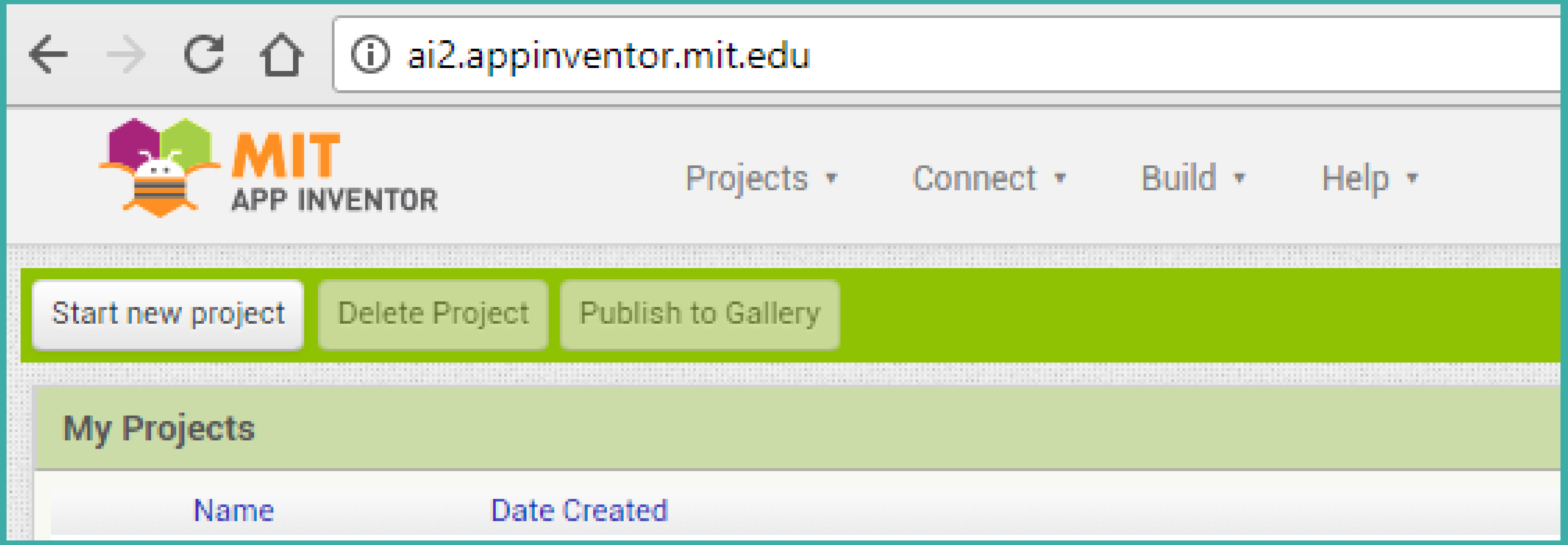
Android phone

Setting up App Inventor

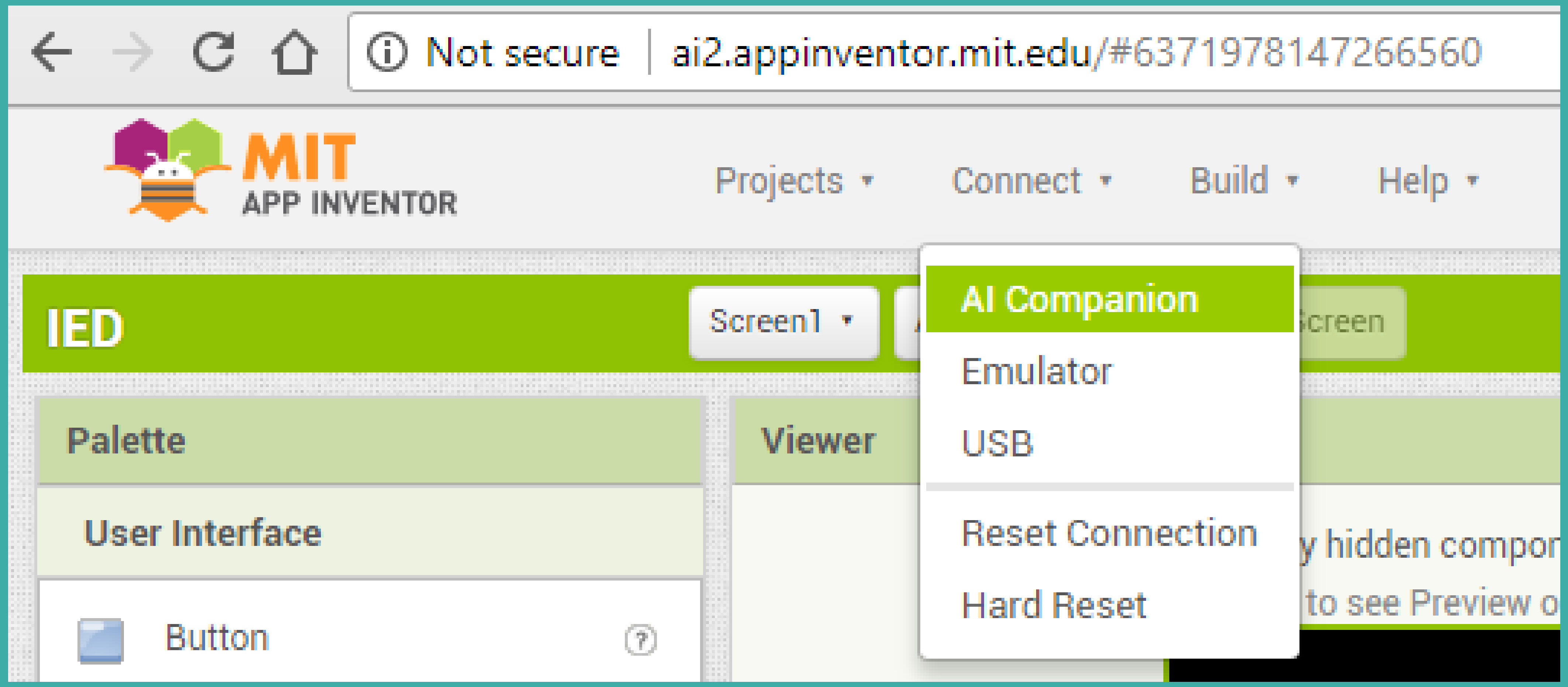
- You'll develop apps on: ai2.appinventor.mit.edu
- Install the App Inventor Companion App (MIT AI2 Companion) on your device for live testing
- Both Android device & laptop should be connected to same WiFi network
- Open projects in App Inventor on the web, open the companion on your device, and you can test your apps as you build them

STEPS:

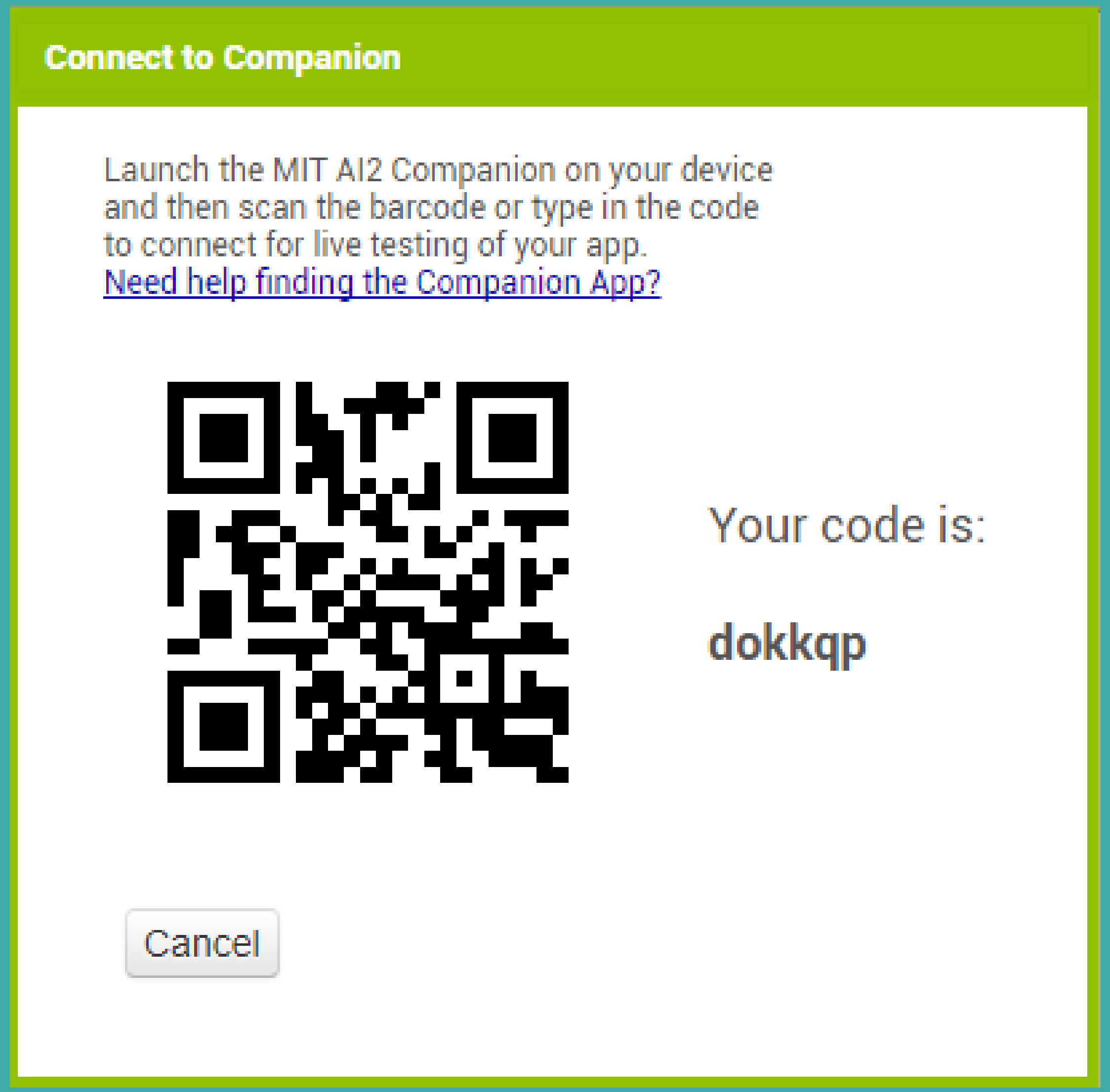
1)Click on “Start New Project”



2) Choose "Connect" and "AI Companion" from the top menu



3) A dialog with a QR code will appear on your PC screen.



4) Open MIT AI2 App on your device & scan this QR code or type the 6 digit code which is appearing on your PC screen.

5) Now you can see the app you are building on your device.

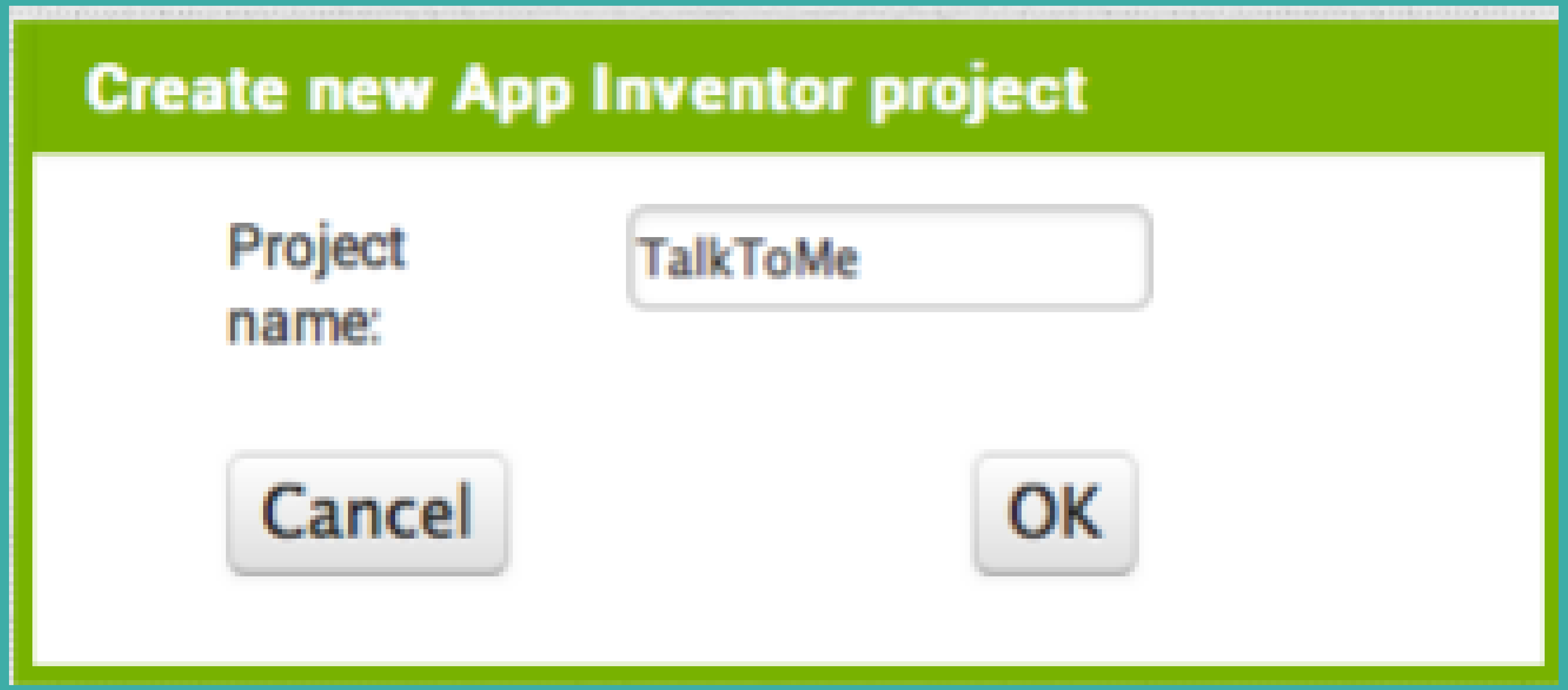
Activity 1

Develop Talk to Me App

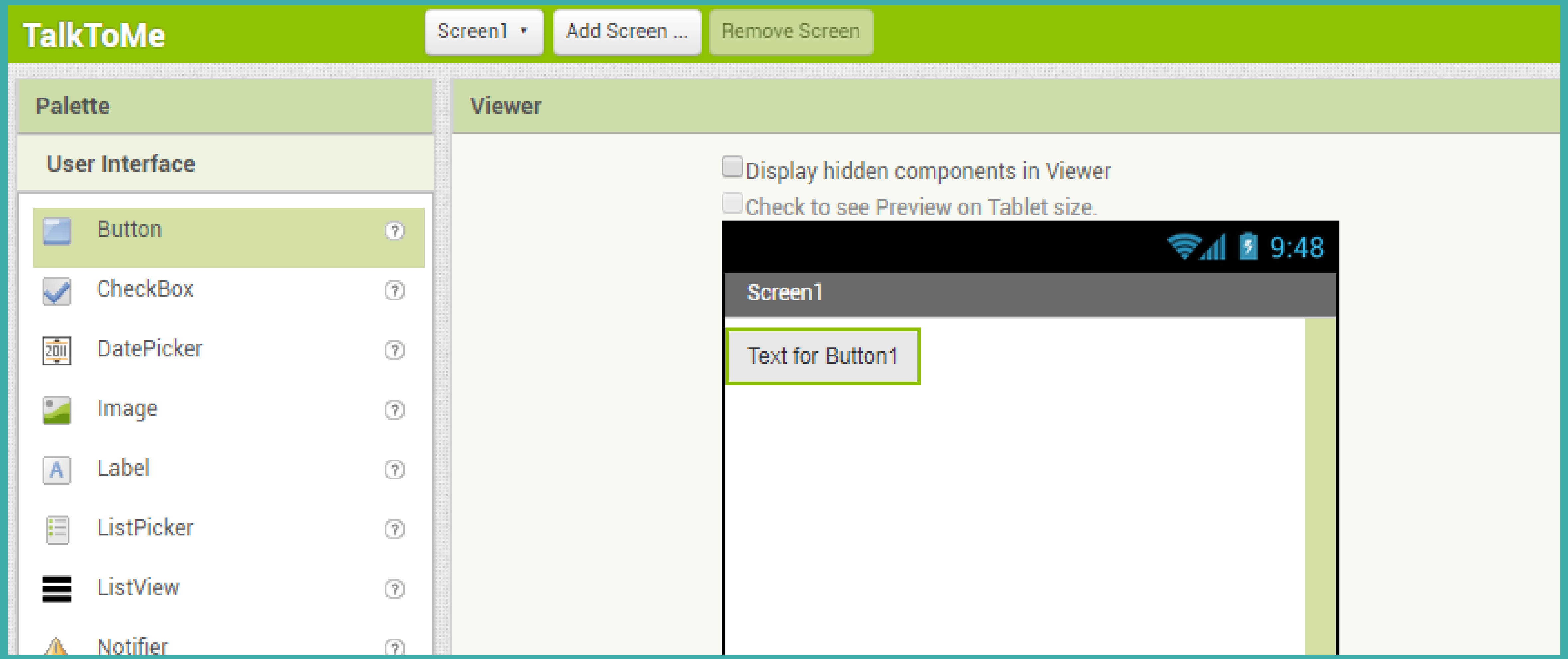
Hear your message when you press the button

Steps:

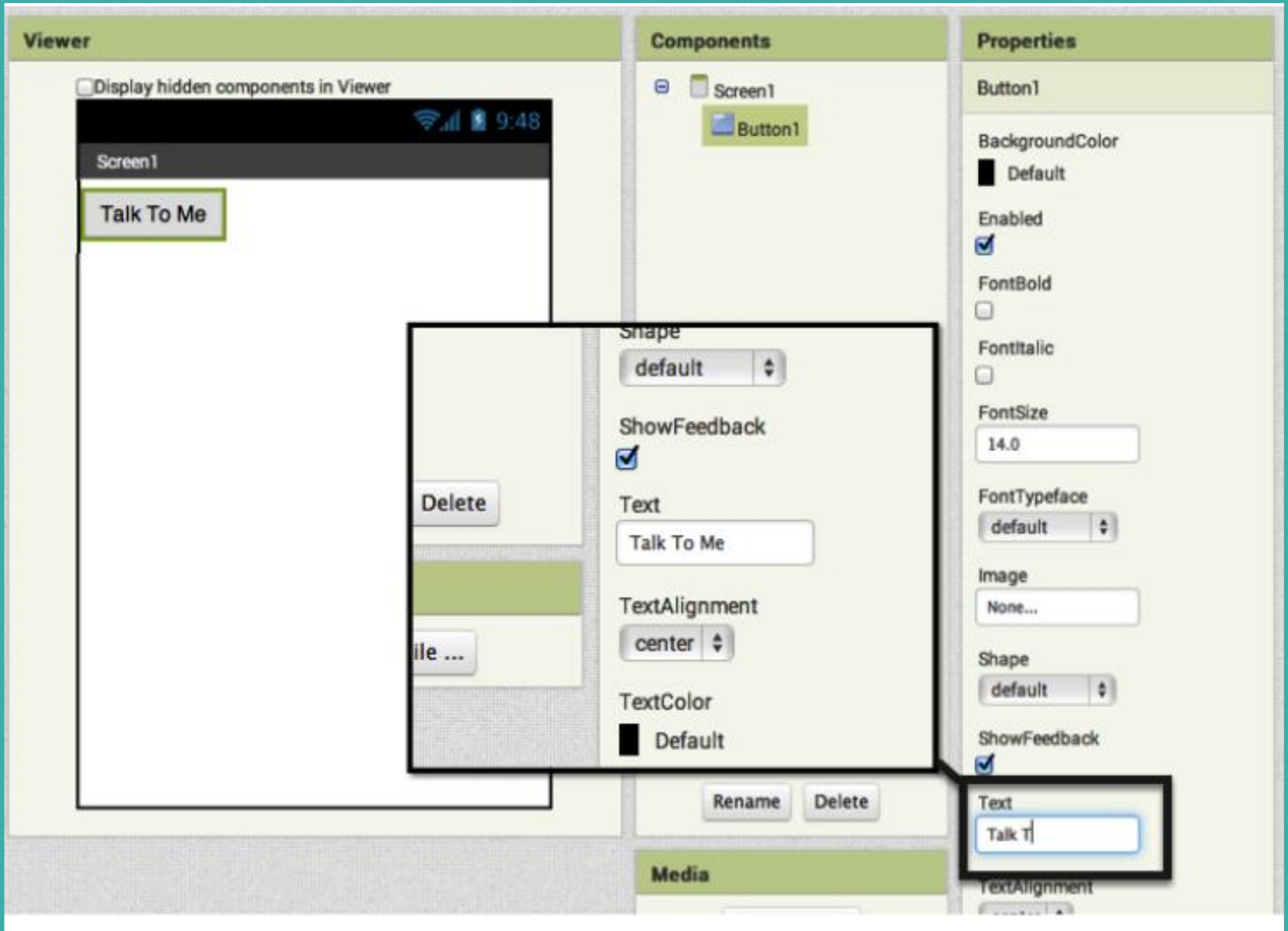
1) Name the project "TalkToMe" with no spaces, only underscores are allowed.



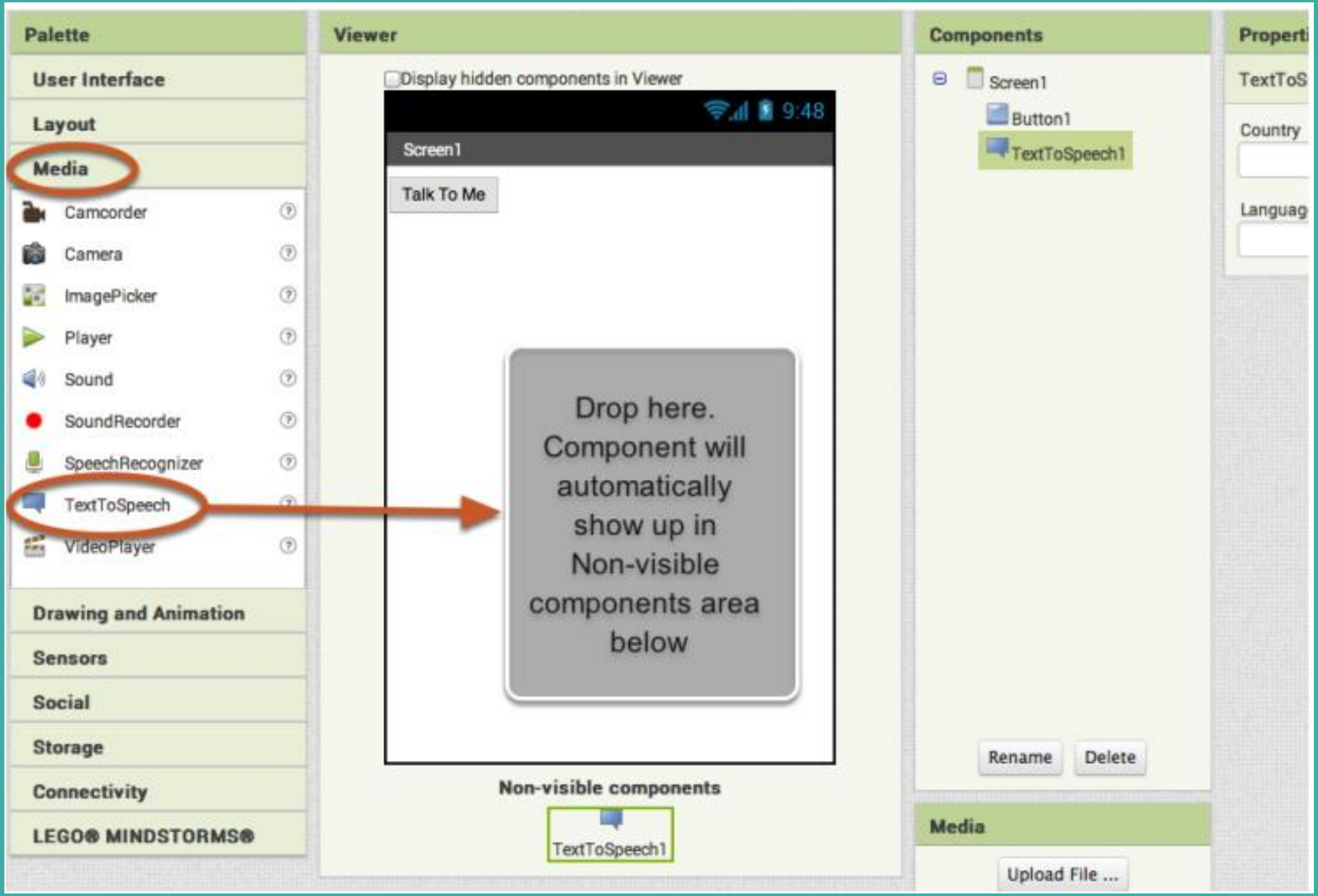
2) Drag 'Button' from left (Palette). Hold and place to 'Screen1' (Viewer).



3) Change the button name by typing 'Talk To Me' in 'Text' space under 'Properties' tab. The same will be reflected on your phone screen as well.

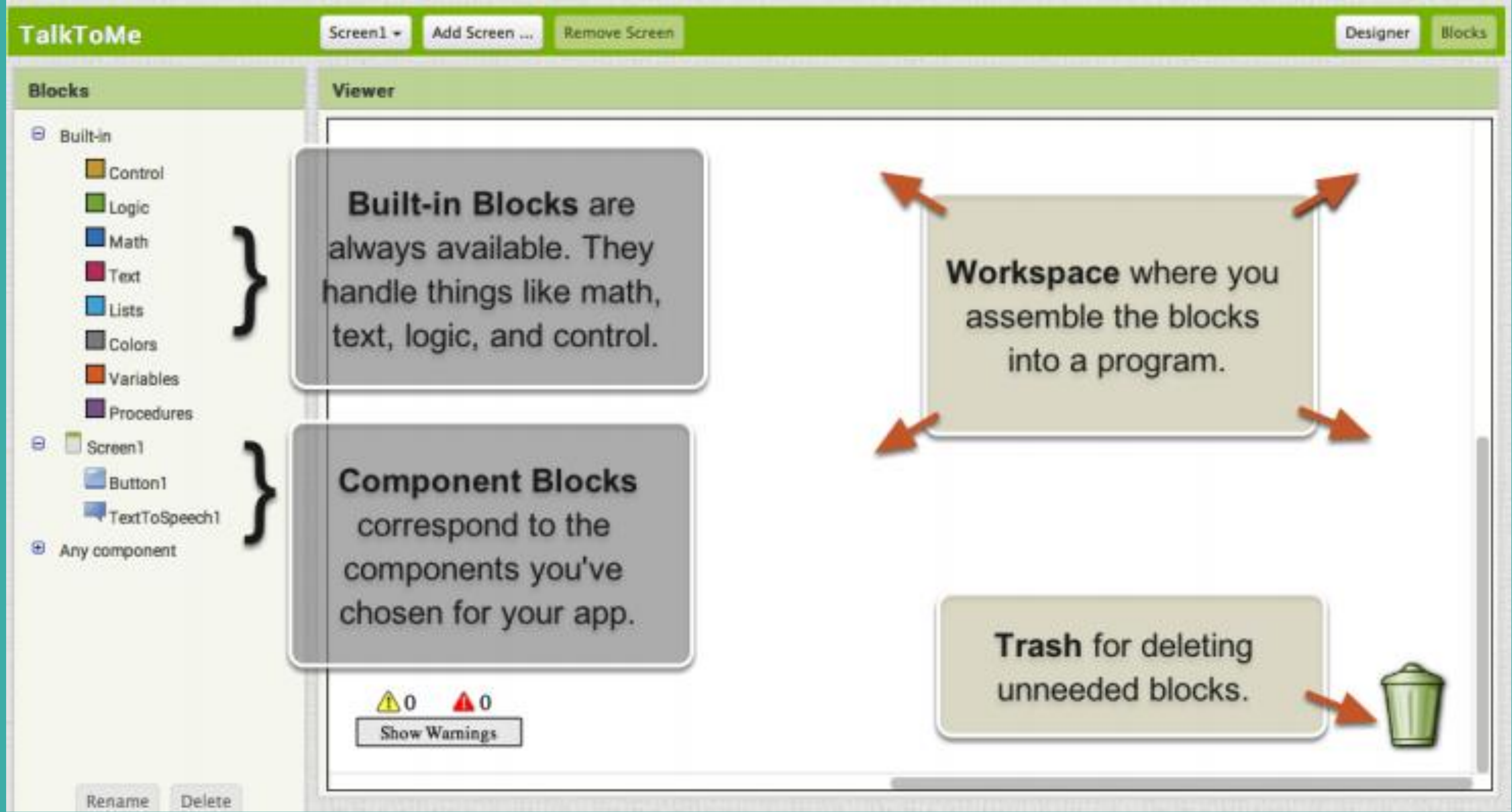


4) Go to Media-> Drag Text To Speech component and drop it on the screen.

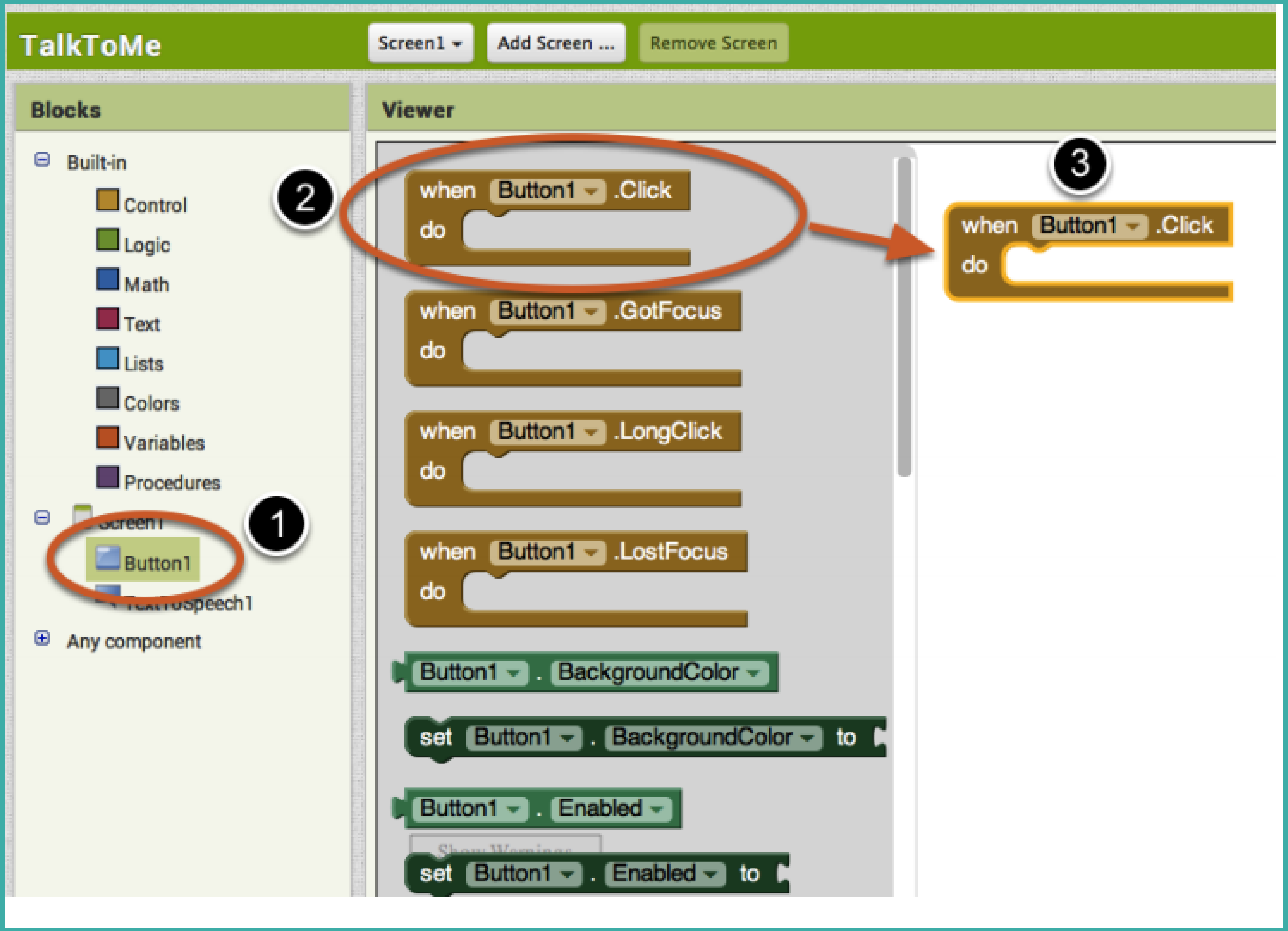


5) It's time to tell your app what to do. Select 'Blocks' on the right top corner. Here you can program the behaviour of your app.

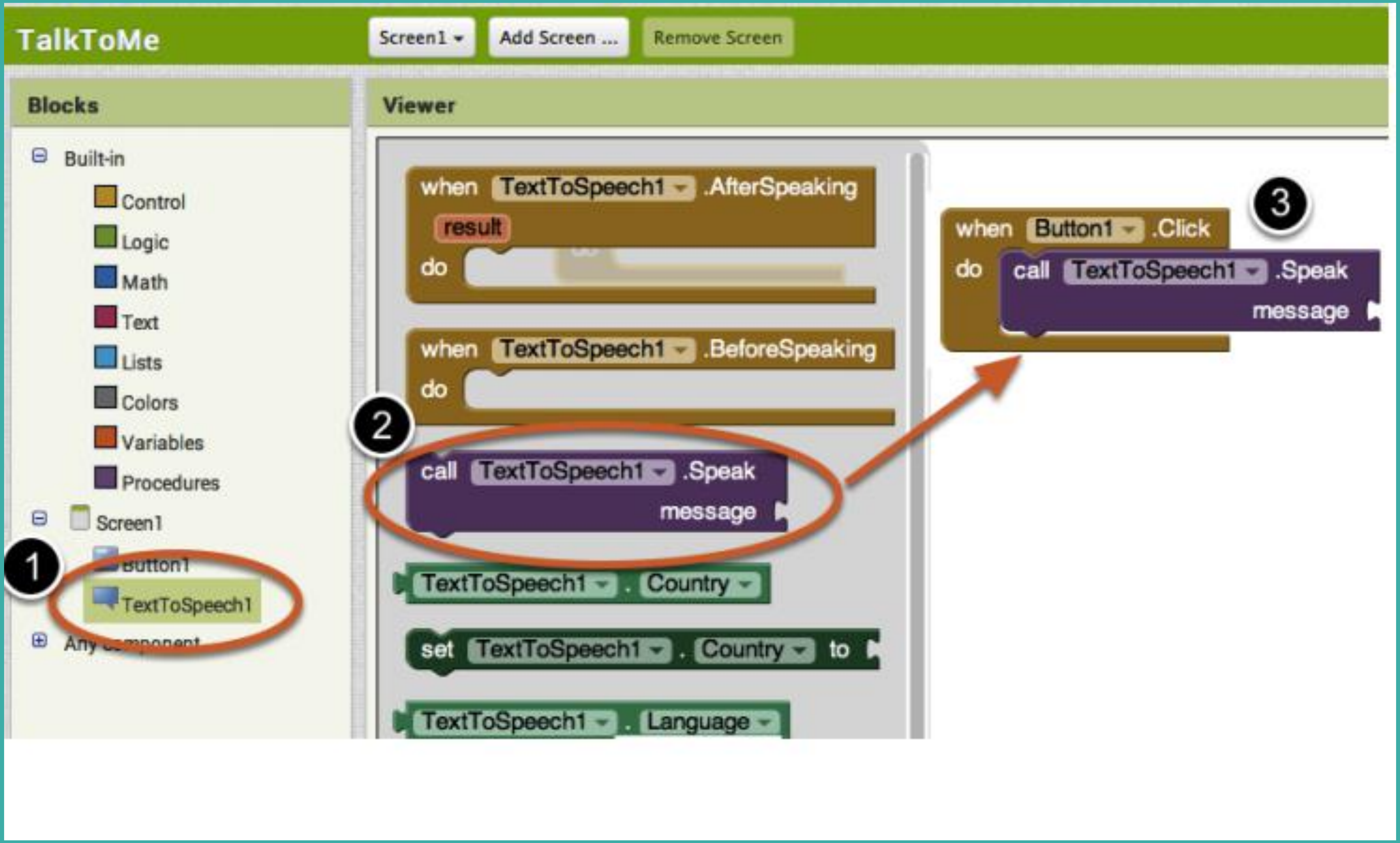
6)There are Built-in blocks that handle things like math, logic, and text.



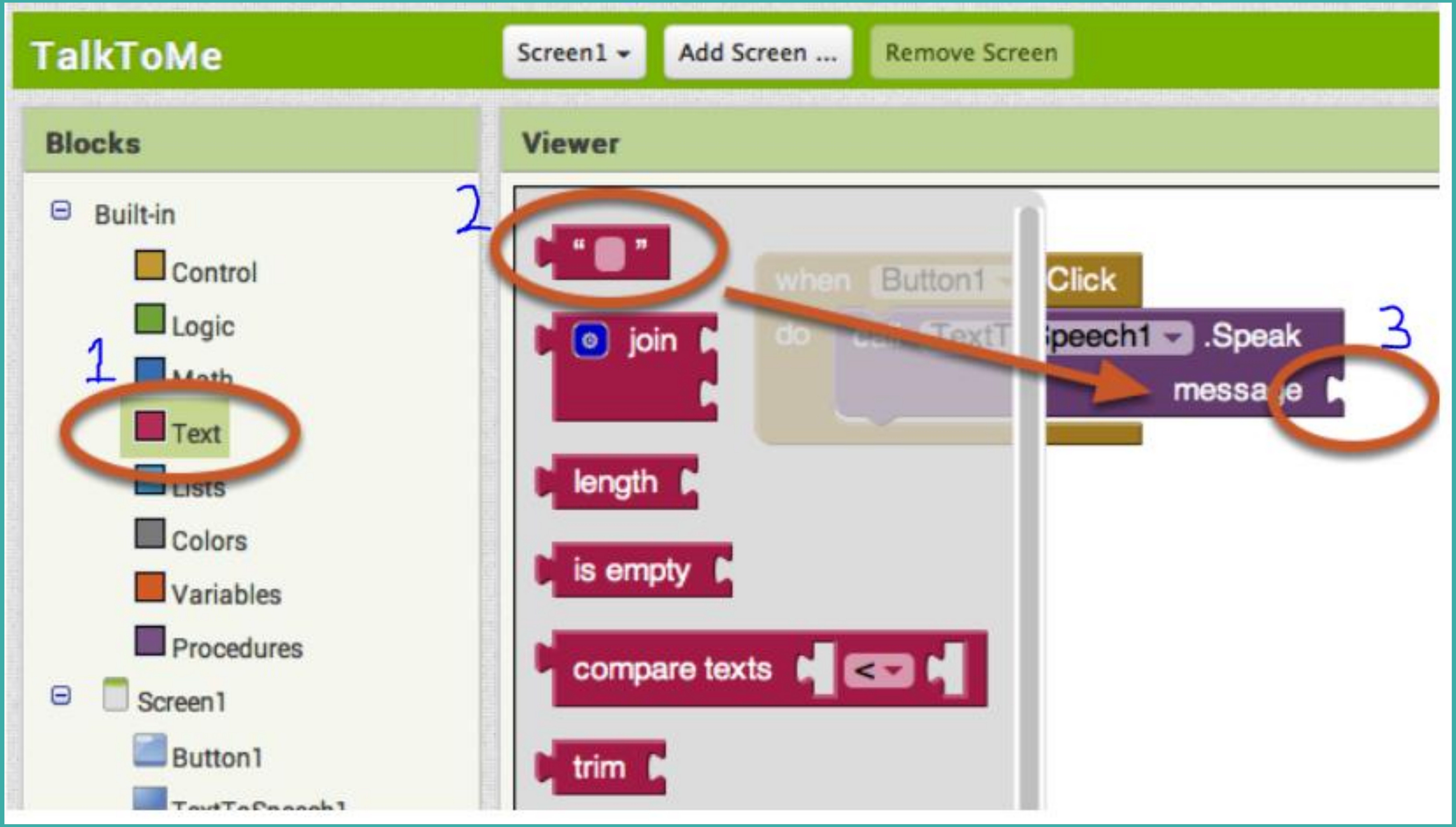
7) Make a Button Click Event. Follow 1 -> 2 -> 3.This block will launch when the button on your app is clicked. It is called an "Event Handler".



8) Program Text to Speech Action. Follow 1 -> 2 -> 3.This is the block that will make the phone speak. Because it is inside the Button.Click, it will run when the button on your app is clicked.

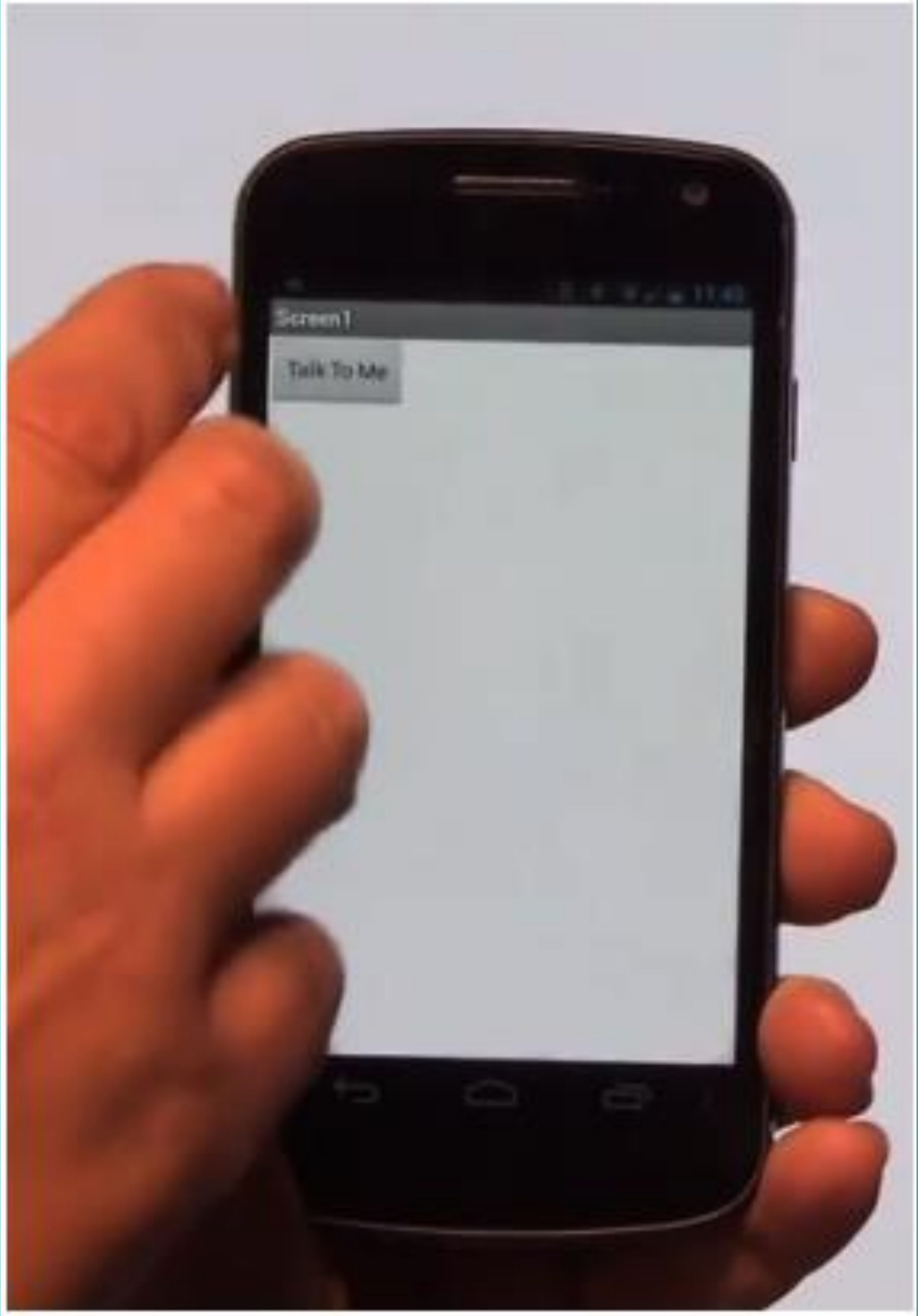


9) Type the message. Follow 1 -> 2 -> 3.Now you need to tell the TextToSpeech.Speak block what to say.



Testing Time!!!

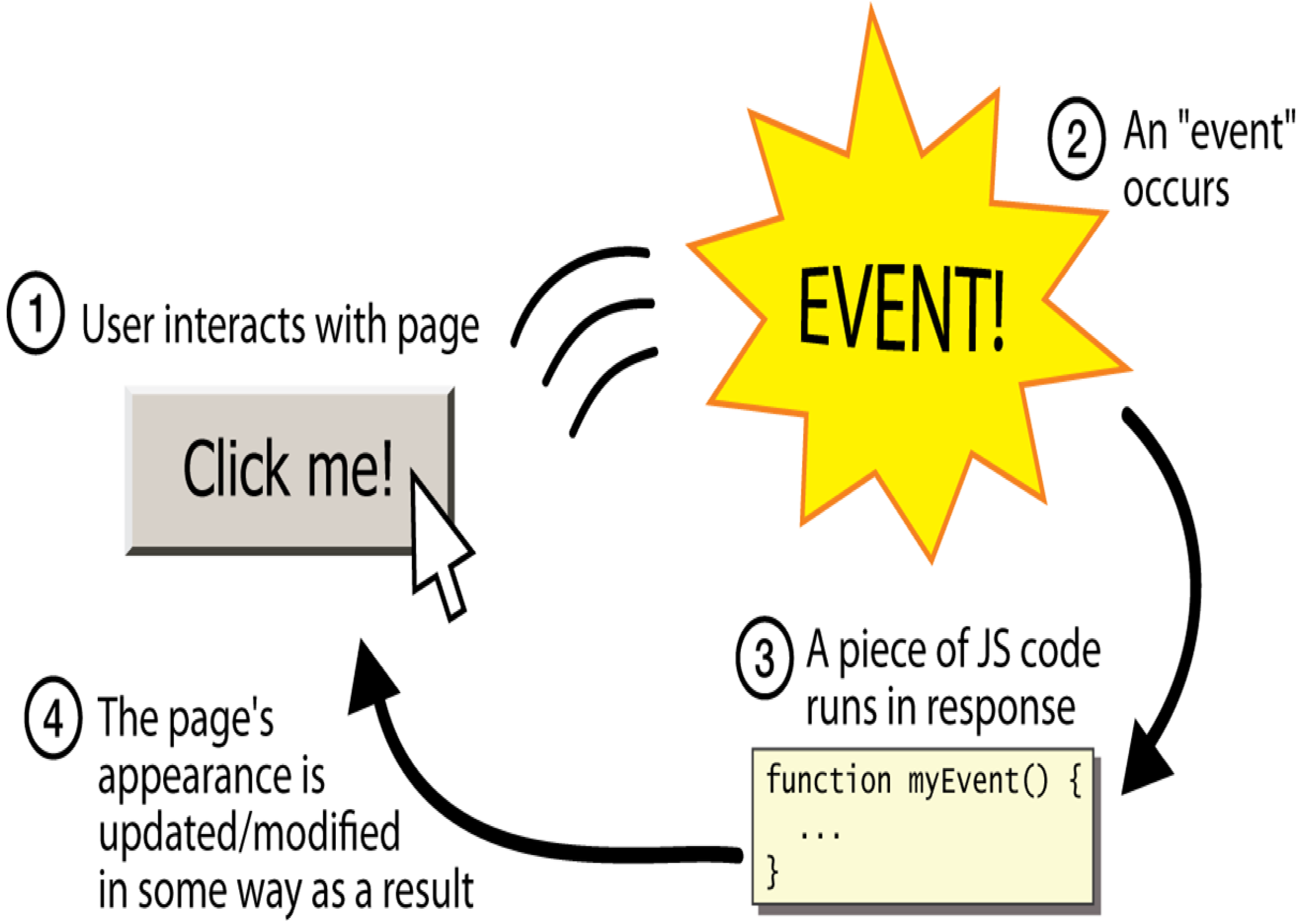
Click on the button on your phone screen, you can hear your typed message...



Event Based Programming

A programming paradigm in which the flow of **program execution** is determined by events - for example a user action such as a **mouse click**, key press, or a message from the operating system or another program.

Android is “Event based programming “



Try out remaining activities from the attached link

https://bit.ly/1f28kWg

Scan me